FHSS Transceiver with Hop Synchronization and Partial-Band Interference

Wireless Communication Project Report

Wireless Communication Class Project

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1 Abstract

This project designs and evaluates a frequency-hopping spread-spectrum (FHSS) transceiver in base MATLAB. A framed BPSK payload is transmitted over eight pseudo-random carrier offsets while a jammer occupies only part of the hopping band. The receiver performs preamble-assisted hop timing and hopping-code acquisition, dehopping, coherent BPSK detection, frame synchronization, and CRC-16 checking. Reliability is measured using payload bit error rate (BER), packet error rate (PER), and synchronization probability. Experiments vary jammer occupancy, hop rate, hopping-code mismatch, and receiver hop-chip-time mismatch, then compare FHSS with an equal-power fixed-frequency BPSK link. The complete experiment is deterministic and requires no MATLAB toolboxes.

2 Objectives

The work addresses four objectives:

1. Synthesize and visualize a reproducible framed FHSS waveform.
2. Model AWGN and a configurable partial-band interferer.
3. Recover the payload through hop and frame synchronization, dehopping, BPSK demodulation, and CRC validation.
4. Quantify BER/PER and make an engineering recommendation for hop-set size and hop rate.

The implementation is a complex low-intermediate-frequency equivalent of a nominal 2.4 GHz radio. Directly sampling a 2.4 GHz carrier is unnecessary because frequency translation preserves the modulation and interference relationships being studied.

3 System Design

3.1 Transmitter and receiver architecture

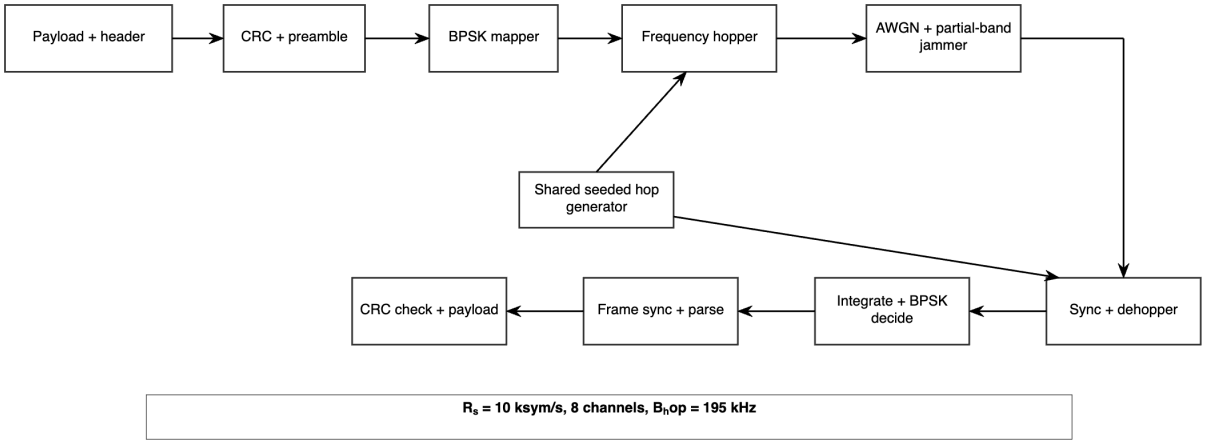


Figure 1: FHSS transmitter, channel, and receiver processing chain.

The transmitter appends a fixed synchronization word and CRC to each payload, maps the bits to BPSK symbols, and mixes each hop-chip interval to a carrier selected by a shared pseudo-random generator. The receiver evaluates timing and hopping-code candidates with the known preamble, reconstructs the winning carrier schedule, dehops the samples, and uses integrate-and-dump symbol decisions. A frame is accepted only if preamble detection and CRC validation succeed.

3.2 Frame and modulation

The fixed-length 320-bit frame is shown in Table 1. The CRC covers the header and payload but not the preamble.

Table 1: Frame format

Field	Length	Purpose
Synchronization word	32 bits	Hexadecimal `0x1ACFFC1D`
Frame counter	16 bits	Packet identification
Payload	256 bits	BER measurement data
CRC-16	16 bits	Packet acceptance

CRC-16/CCITT-FALSE uses polynomial $0x1021$, initial state $0xFFFF$, no input/output reflection, and no final XOR. Its standard `123456789` check value is $0x29B1$.

BPSK mapping is

$$a_k = 1 - 2b_k,$$

where $b_k \in \{0, 1\}$ and $a_k \in \{+1, -1\}$. Rectangular pulse shaping gives 50 samples/symbol at $R_s = 10$ ksymbol/s and $F_s = 500$ ksample/s. Transmit power is normalized to unity.

3.3 Hopping design and bandwidth

The eight low-IF carrier offsets are

$$[-87.5, -62.5, -37.5, -12.5, 12.5, 37.5, 62.5, 87.5] \text{ kHz}.$$

They are separated by $\Delta f = 25$ kHz. Rectangular BPSK has an approximate null-to-null occupied width $B_{\text{ch}} = 2R_s = 20$ kHz, leaving a 5 kHz separation margin. The total hopping bandwidth is

$$B_{\text{hop}} = (N - 1)\Delta f + B_{\text{ch}} = (8 - 1)(25) + 20 = 195 \text{ kHz}.$$

Here, a **hop chip** is one interval spent on a single hop frequency. Its duration, T_{hc}, is also called the hop dwell time. It is not a BPSK symbol time: each hop chip contains N_s BPSK symbols. For N_s symbols per hop,

$$T_{\text{hc}} = \frac{N_s}{R_s}, \qquad R_h = \frac{1}{T_{\text{hc}}} = \frac{R_s}{N_s}.$$

Table 2: Hopping configurations

Symbols/hop	Hop chip time	Hop rate	Hops/frame
16	1.6 ms	625 hops/s	20
8	0.8 ms	1,250 hops/s	40
4	0.4 ms	2,500 hops/s	80

The hopping generator uses a fixed Mersenne Twister seed. Successive random permutations of all eight channels ensure balanced use while retaining a reproducible pseudo-random order.

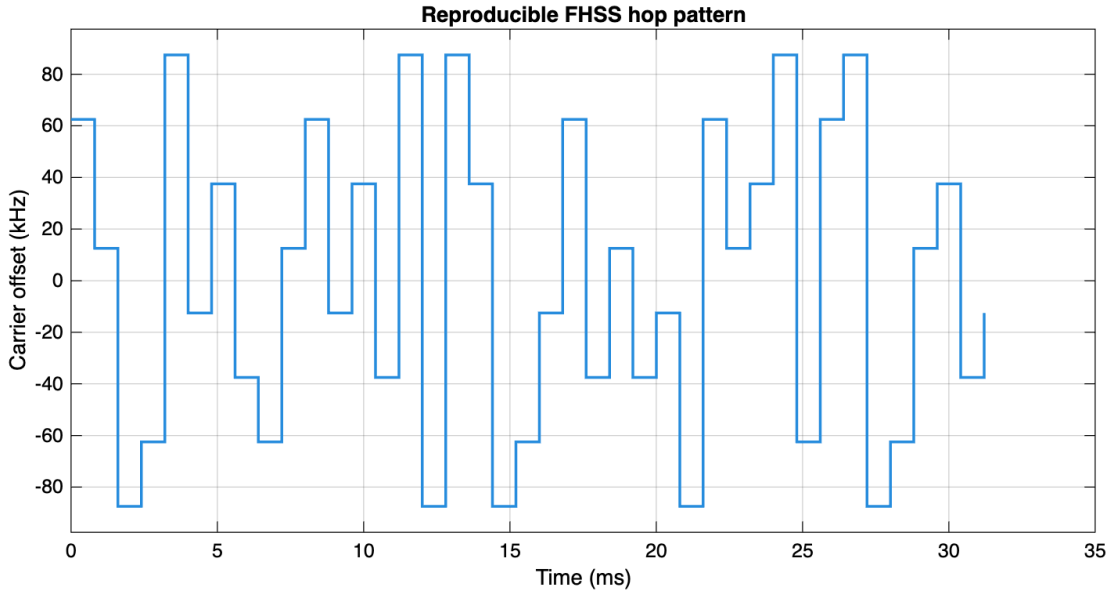


Figure 2: Baseline pseudo-random carrier sequence.

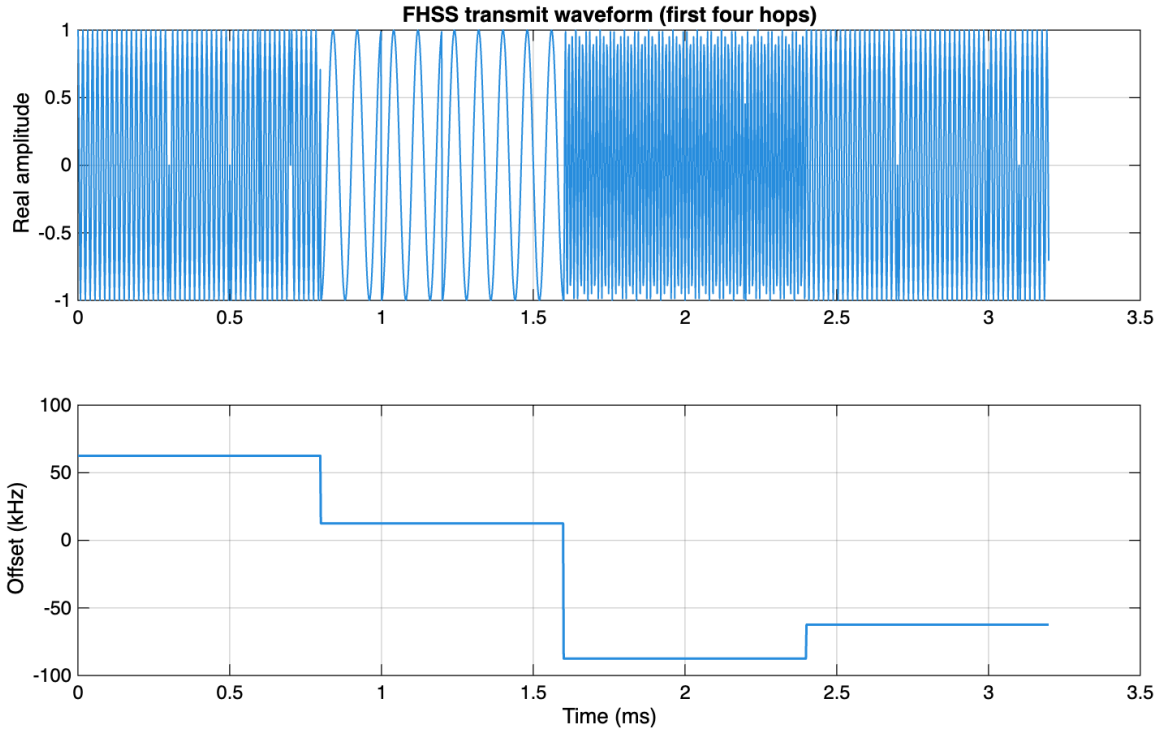


Figure 3: Real transmit waveform and instantaneous hop offset.

4 Channel and Interference Model

Let $\gamma_b = E_b/N_0$ denote the **linear** energy-per-bit to noise-density ratio. Its decibel representation is $(E_b/N_0)_{\text{dB}} = 10\log_{10}(\gamma_b)$. For unit BPSK amplitude and L samples/symbol, the complex AWGN sample-noise variance is

$$\sigma_n^2 = \frac{L}{\gamma_b}.$$

This produces a real decision-noise variance of $1/(2\gamma_b)$ after integrate-and-dump detection. Without interference, the expected coherent BPSK result is

$$P_b = Q\left(\sqrt{2\gamma_b}\right) = \frac{1}{2}\operatorname{erfc}(\sqrt{\gamma_b})\,.$$

The partial-band jammer occupies a contiguous central subset of the hop channels. Independent complex Gaussian symbols are rectangularly shaped and translated to every occupied jammer channel. Aggregate jammer power is fixed at $J/S = -10$ dB and divided equally between occupied channels. Consequently, increasing occupancy changes both the probability of a hop collision and jammer power per occupied channel; it does not silently increase total jammer power.

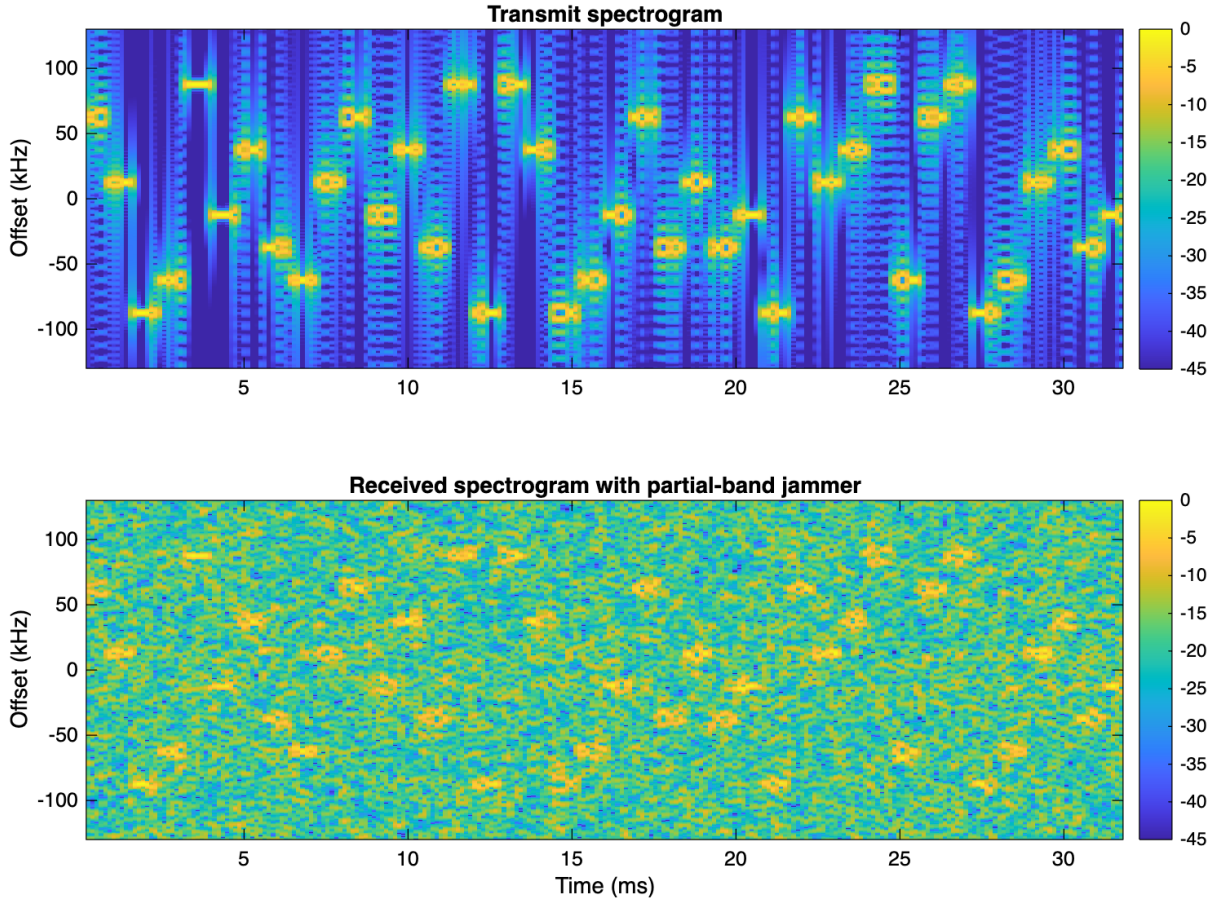


Figure 4: Transmit and received spectrograms for 25% jammer occupancy.

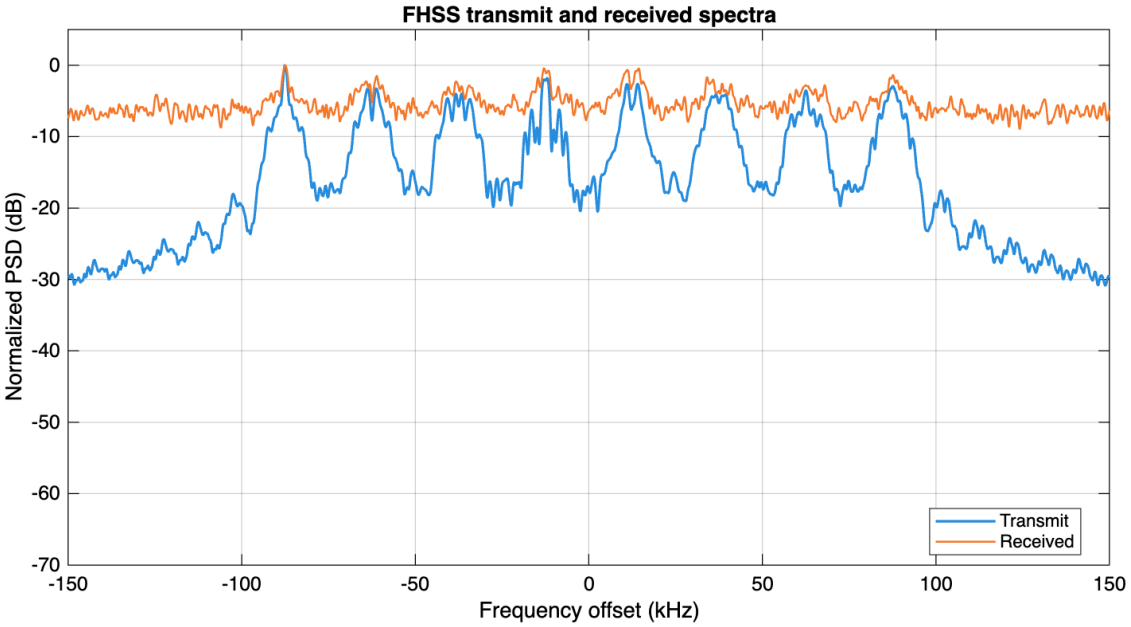


Figure 5: Normalized transmit and received power spectra.

5 Receiver Synchronization

The receiver knows the hop set, seed, symbol rate, and nominal hop chip time but not the exact packet boundary or hopping-code phase. It searches code offsets from -8 through $+8$ hops and evaluates every sample timing within one hop chip. The exhaustive timing search avoids carrier-dependent phase ambiguity that can make a symbol-spaced coarse search select the wrong refinement region.

For a candidate soft preamble vector $\mathbf{y}$ and known bipolar preamble $\mathbf{p}$, the acquisition metric is normalized correlation:

$$M = \frac{\mathbf{y}^T\mathbf{p}}{\sqrt{(\mathbf{y}^T\mathbf{y})(\mathbf{p}^T\mathbf{p})}}.$$

Acquisition requires $M \geq 0.70$ and at least 28 of 32 correct hard preamble decisions. After dehopping, the receiver performs another Hamming-distance preamble search, extracts the frame, and verifies CRC. A synchronization failure or CRC failure counts as a packet error. BER is calculated only from the 256 payload bits.

Deliberate impairment tests force a one-hop sequence offset or receiver hop-chip-time errors from -0.5% through $+0.5\%$. Fractional receiver hop boundaries are accumulated before rounding so that the implementation does not add a systematic per-hop rounding bias.

6 Experiment Method

The BER/PER grid uses $E_b/N_0 = 0 : 2 : 14$ dB, jammer occupancies of 0%, 12.5%, 25%, and 50%, and all three hop rates. Every operating point uses at least 250 packets and stops after 100 packet errors or 2,000 packets. Wilson 95% confidence intervals quantify Monte Carlo uncertainty.

The aligned-link grid assumes matched hop rates. The mismatch study varies receiver hop chip time from -0.5% to +0.5%, with finer steps near zero, at $E_b/N_0 = 10$ dB, 25% occupancy and $J/S = -10$ dB. BER/PER use 200 packets per point with acquisition disabled and packet timing and initial hop alignment supplied.

The frame-synchronization panel uses a separate acquisition-enabled sweep with 50 packets per point, sharing the first 50 packet seeds with the BER/PER case at each setting. Acquisition searches at the nominal hop duration and does not estimate or correct hop-duration error. An additional zero-mismatch acquisition study uses 50 packets per hop rate.

The complete sample-domain waveform is used for the aligned-link BER/PER grid, synchronization, code-phase error and hop-rate mismatch. Acquisition is disabled for the aligned-link grid and deliberate one-hop offset study; this preserves the imposed misalignment in the latter. A fast symbol-domain model is retained for the fixed-frequency decision-level comparison. Selected operating points compare that abstraction with the waveform model using known timing to isolate channel-model differences.

The fixed-frequency comparison preserves BPSK symbol rate and average transmit power. It includes fixed jammed and clean carriers and a fixed channel chosen uniformly per packet, preventing a single favorable carrier choice from being mistaken for general fixed-link performance.

7 Results and Analysis

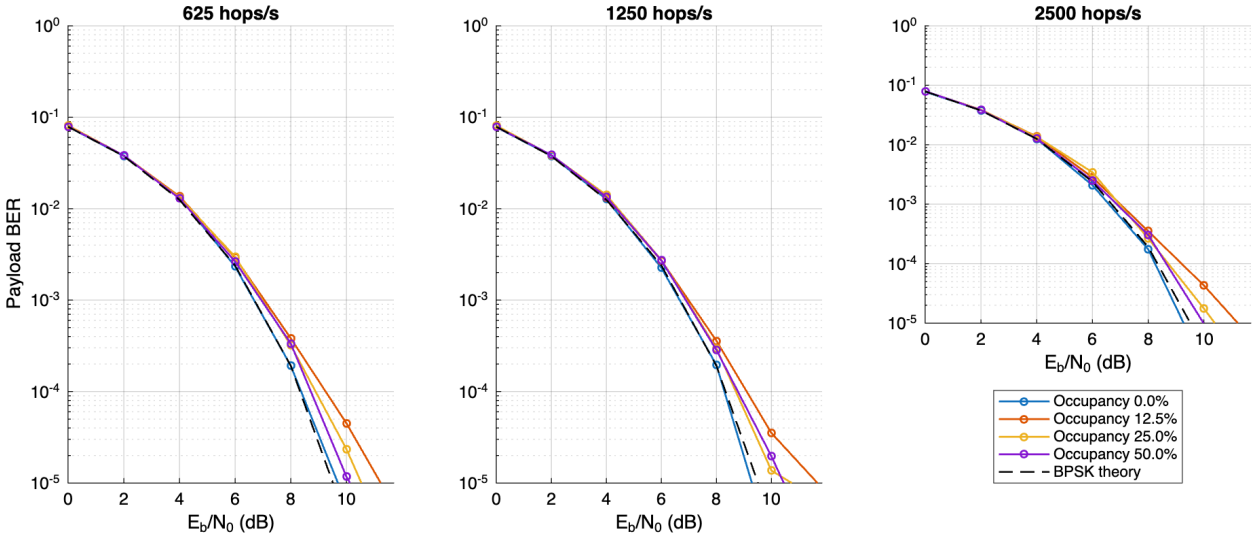


Figure 6: BER across jammer occupancy and hop rate. The dashed reference is theoretical jammer-free BPSK.

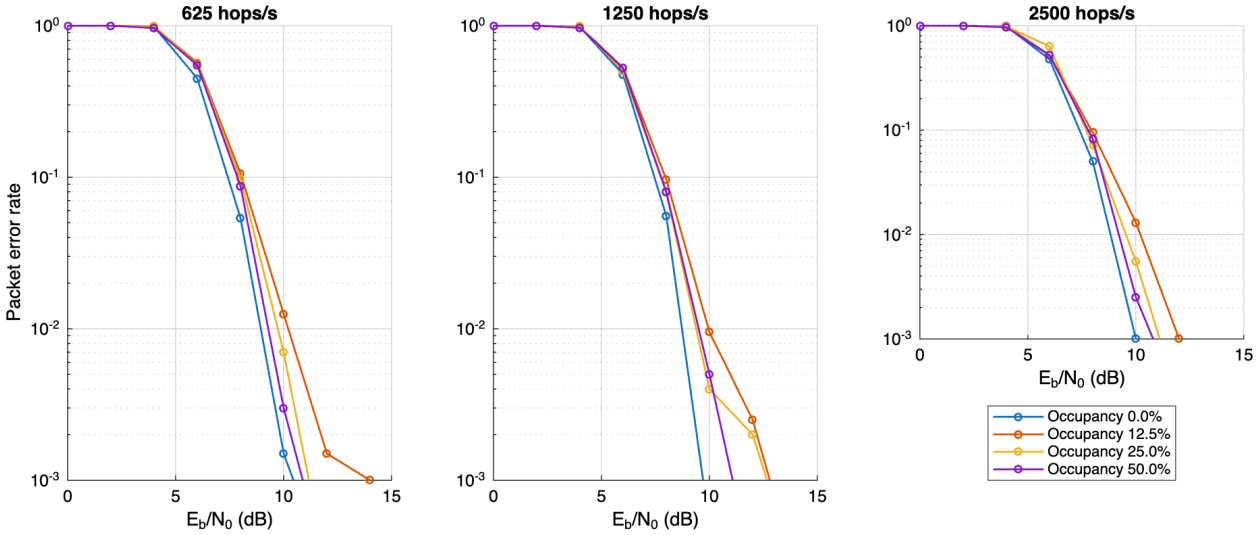


Figure 7: PER across jammer occupancy and hop rate.

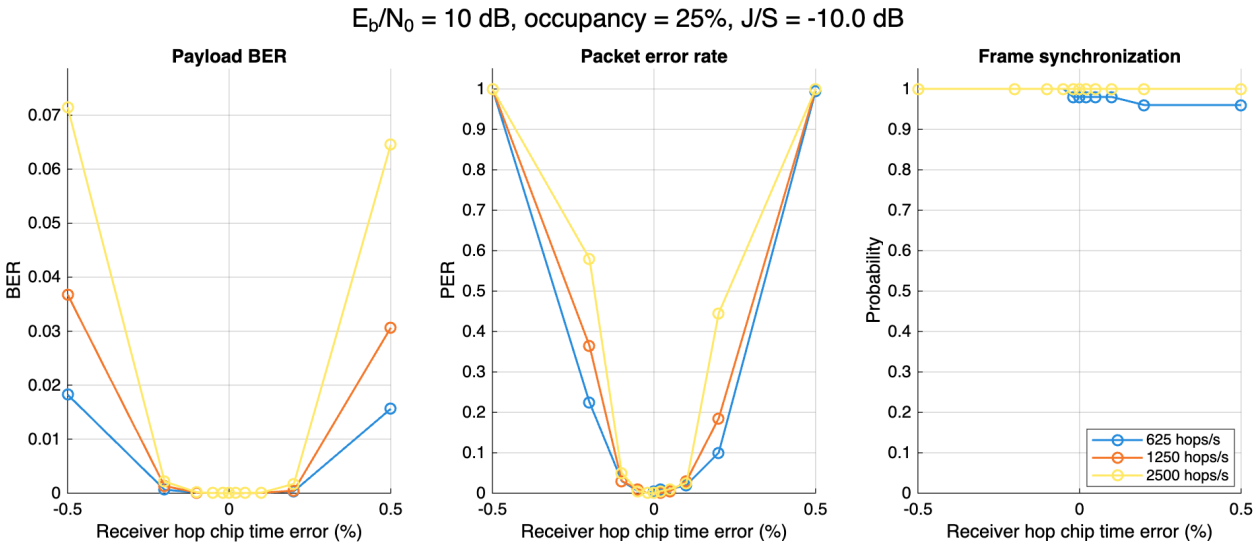


Figure 8: Receiver hop-chip-time mismatch: BER/PER with acquisition disabled; frame synchronization with acquisition enabled.

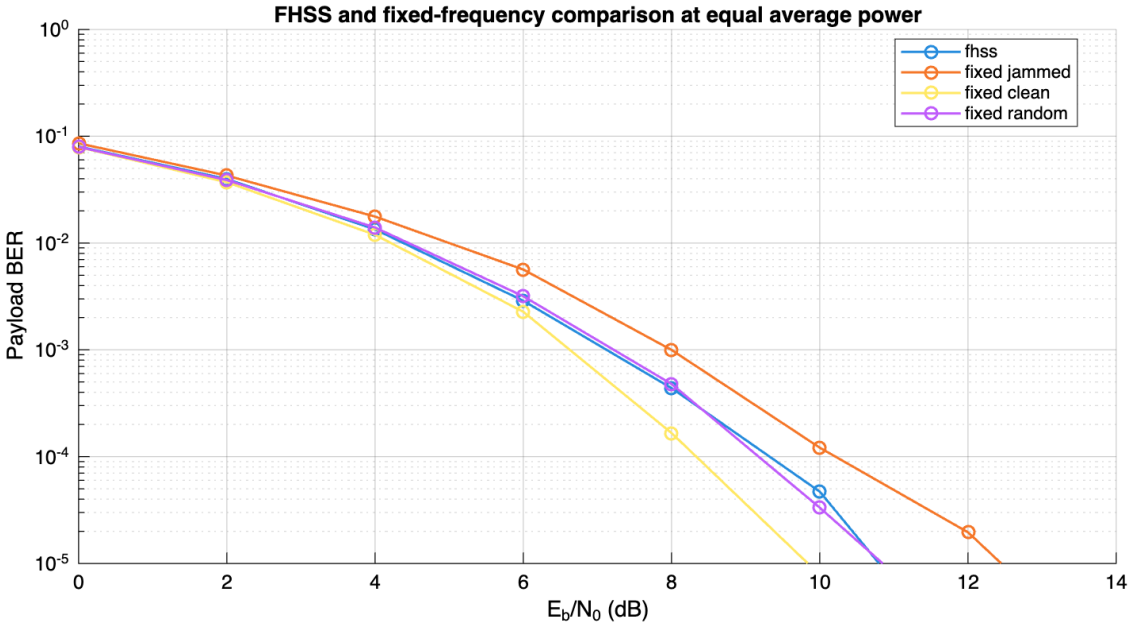


Figure 9: FHSS and fixed-frequency BER at equal average transmit power.

The baseline at 10 dB used 1250 hops/s. Table 3 reports the measured payload BER and PER as jammer occupancy increased.

Jammer occupancy	BER	95% BER interval	PER	95% PER interval
0.0%	1.953e-06	[3.448e-07, 1.106e-05]	0.0005	[8.827e-05, 0.002827]
12.5%	3.516e-05	[2.224e-05, 5.558e-05]	0.0095	[0.00609, 0.01479]
25.0%	1.367e-05	[6.623e-06, 2.822e-05]	0.004	[0.002028, 0.007873]
50.0%	1.953e-05	[1.061e-05, 3.596e-05]	0.005	[0.002718, 0.00918]

The provisional engineering recommendation is an eight-channel hop set at **1250 hops/s**, corresponding to **0.80 ms per hop**. The eight configured channels occupy a nominal 195 kHz span with 25 kHz spacing. The rate selection considers acquired packet errors as well as known-timing performance; preamble recognition alone does not establish correct hop alignment. Larger acquisition-enabled trials are needed to confirm the choice.

Table 4 compares FHSS with fixed-frequency alternatives at equal average transmit power.

Link	BER	PER	Packets
fhss	4.688e-05	0.0125	2000
fixed jammed	0.0001211	0.0325	2000
fixed clean	7.813e-06	0.002	2000
fixed random	3.32e-05	0.0105	2000

The hop-chip-time mismatch BER/PER study uses 200 packets per point with acquisition disabled and supplied timing and initial code phase. The frame-synchronization panel uses a separate acquisition-enabled sweep with 50 packets per point.

Hop rate	Hop chip time error	Packet errors / packets	PER
625 hops/s	-0.1%	6 / 200	0.030
625 hops/s	+0.0%	1 / 200	0.005
625 hops/s	+0.1%	4 / 200	0.020
1250 hops/s	-0.1%	6 / 200	0.030
1250 hops/s	+0.0%	0 / 200	0.000
1250 hops/s	+0.1%	6 / 200	0.030
2500 hops/s	-0.1%	10 / 200	0.050
2500 hops/s	+0.0%	0 / 200	0.000
2500 hops/s	+0.1%	5 / 200	0.025

Table 5 contrasts forced one-hop misalignment with preamble-assisted acquisition.

Hop rate	Offset BER	Offset PER	Acquired sync	Acquired PER
625 hops/s	0.4961	1	1.000	0.220
1250 hops/s	0.491	1	1.000	0.020
2500 hops/s	0.4941	1	1.000	0.000

The acquisition columns use 50 packets per rate in a separate zero-mismatch experiment; the forced-offset columns use 200 packets per rate. Acquired synchronization records passing the preamble checks and can accept a wrong hopping-code phase. Acquired PER includes CRC failures and therefore exposes payload failures hidden by that synchronization metric.

The symbol-domain and full-waveform BER confidence intervals overlapped in 3 of 3 validation cases. The jammer cases reveal the expected effect of asynchronous jammer symbol boundaries and adjacent-channel leakage. Accordingly, the main BER/PER grid, mismatch, and synchronization results use the full sample-domain waveform; the symbol model is retained only for the fixed-link decision-level comparison.

The occupancy curves must be interpreted with the jammer normalization in mind. A narrow jammer concentrates the fixed total power into fewer channels, producing severe but less frequent hop collisions. A wider jammer collides with more hops but has less power in each occupied channel. PER is often more sensitive than BER because a single corrupted payload or CRC bit rejects the entire frame.

Receiver hop-chip-time mismatch accumulates timing error across the packet, so frame detection can succeed even when later bits are corrupted. Matched hop-rate changes do not necessarily create a large BER improvement in an ideal instantaneous-switching channel. Its main value is the time scale over which interference errors are distributed.

8 Engineering Recommendation

Use all eight configured hopping channels within the nominal 195 kHz span. With a fixed number of jammed channels, using more available channels reduces the fraction of the hop set directly affected. This choice applies to the specified channel plan; other hop-set sizes were not experimentally optimized.

The provisional rate reported in Results and Analysis is selected by screening acquired packet error rate, rather than preamble recognition alone. Retain rates whose acquired PER is within five percentage points of the best observed value, then choose the lowest known-timing PER at 10 dB and 25% jammer occupancy. The five-percentage-point tolerance is an engineering heuristic, not a statistical equivalence test. Overlapping confidence intervals do not establish equal performance.

In the current measurements, acquired PER is 22%, 2%, and 0% at 625, 1250, and 2500 hops/s, respectively, using only 50 packets per rate. The 32-bit preamble spans two, four, and eight hops, respectively. At 625 hops/s, distinct code positions can share the first two channels, causing a wrong alignment to pass the preamble checks. The zero observed failures at 2500 hops/s do not establish superiority over 1250 hops/s. More acquisition-enabled packets and a preamble or candidate-checking method that resolves ambiguous hop alignment are needed before a final design choice. Hardware switching cost and synthesizer settling are not modeled.

9 Limitations

This is an idealized simulation. It excludes multipath fading, Doppler, independent symbol-clock error, nonlinear amplifiers, synthesizer settling time, carrier phase estimation, forward-error correction, and SDR hardware. Carrier phase and frequency are assumed known after correct dehopping. These exclusions isolate hopping synchronization and partial-band interference, which are the stated project objectives.

The symbol model assumes ideal channel isolation and does not perform acquisition. Its interference results are an approximation, rather than a substitute for waveform measurements. BER Wilson intervals assume independent bit errors; jammer errors can cluster within hops, so those intervals are approximate. PER intervals treat packets as independent trials. Zero measured error rates are drawn at half an event divided by the trial count on logarithmic plots; the CSV tables retain the actual zero counts.

10 Reproduction

From the project directory, run:

```
/Applications/MATLAB_R2026b.app/bin/matlab -batch "run_all;"
zsh build_report.sh
```

`run_all` performs deterministic tests, writes CSV/MAT evidence, exports every figure as PNG and vector PDF, and fills this report with measured values. All random processes use stored seeds.